

1. remaining concepts of hf
 2. full fledged ft pipeline using hf
instruction tuning
custom raw data(docs)
inbuilt instruction tuning data from hf
 3. ft pipeline using unsloth
 4. LORA, QLORA, DORA(assignment)
 5. RLHF, DPO
- 2-3 classes

RAG,MMRAG

LLM,LLMops

LLMops,LLMops

Deepseek-> Chain
Servam-> India
Mistral-> France
OpenAI-> USA

1. Pretraining
2. Instruction tuning->SFT
3. Preference tuning

dataset-> is library
inbuilt dataset--> which is already there over the huggingface
which is upload by different companies, universities, opensource community

custom dataset--> your dataset
your personal
your enterprise

i can create my custom data into hf compatible format and i can upload it over there
so anyone can use it

csv
xlsx
tsv
json

big-data
parquet
rc
orc
binary format of the data

Transformer->training->parallel->token->token_ids->wordembedding->positional_encoding->self_attention->ffnn

token->
1. BPE(byte pair encoding)
2. WordPiece algo

"I love AI"

["I", "love", "AI"]<- BPE, word_piece, sentence_piece, you can create your own custom tokenization method
[40, 1842, 9552]

Model sizes: BERT-Base has 12 Transformer layers(12 encoder layer), 768 hidden units, 12 attention heads (110M parameters). BERT-Large has 24 layers, 1024 hidden units, 16 attention heads (340M parameters).

